

BIN BAY SCREEN HOARDING - WIND CHECK

Ref TW-BINS-001 Rev P01 4 October 2026 | Banksy, for Dal Hopwood

NOT A DESIGN - NOT FOR CONSTRUCTION - NOT A PERMIT TO LOAD

1. WHAT THIS IS

NOT A DESIGN - NOT FOR CONSTRUCTION.

This pack runs the calculation and shows every step. It is not the designer of record.

The design AND its independent check must be produced, checked and OWNED by a competent

temporary works designer to BS 5975-1:2024 / BS 5975-2:2024.

Nothing is erected or loaded against this pack until a permit to load has been issued.

It runs the calculation and shows every step, with the source of every value. It is not competent, qualified or insured, and it cannot select the

design check category, sign a check, or issue a permit to load.

2. THE ASSET AND THE BASIS

Item	Temporary screen hoarding around a commercial bin bay (75 x 225 C24 posts, 18 mm OSB/3)
Height	2.40 m
Line	6.00 m front run, two 1.20 m returns - total 8.40 m
Post centres	2.00 m c/c, 6 posts
Exposure	EXTERNAL - the yard is open to the sky, so a full wind case applies
Site setting	town, assumed. The yard sits between buildings, so 'town' per TWf B.5
Duration	cprob = 1.0 (more than one year)
Ground	average, G = 390 kN/m2 - assumed, see open inputs

3. THE WORKING - STEP, VALUE, SOURCE

Swind = Twind x vb,map x (1 + A/1000)	25.50 m/s	BS 5975-2 simplified method; A = 20 m site altitude
cprob	1.00	TWf2012:01 B.4 - hoarding up more than a year
Cef	1.174	TWf2012:01 Table B.1 - town, 2.5 m high, 2 km from the sea
qp = 0.613 x cprob^2 x Cef x Swind^2	0.468 kN/m2	TWf2012:01 Eq B.3
l/h = 8.40 / 2.40	3.50	TOTAL_RUN 8.40 m, H 2.40 m
cp,net, Zone B, interpolated on l/h	1.50	TWf2012:01 Table B.2: 1.4 at l/h<=3, 1.8 at l/h=5
Fw = qp x Aref x cp,net x eta	1.685 kN/m run	TWf2012:01 Eq B.1, Aref = 2.40 m2/m, eta = 1.0
Mo = Fw x h/2 (max wind)	2.022 kNm/m run	TWf2012:01 s4.5 - note Fw is already the resultant, so h/2, not w h^2/2
Mo LC2 (working wind + notional 0.74 kN/m at 1.2 m)	1.752 kNm/m run	TWf2012:01 s4.2.1 and Table 2
DESIGN Mo	2.022 kNm/m run	the greater of LC1 and LC2
Mo per post at 2.00 m c/c	4.043 kNm	Mo x spacing

Permissible moment, 75 x 225 C24	6.294 kNm	TWf2012:01 Table C1
Bending utilisation	64.2 %	Mo per post / permissible
Qo per post at 2.00 m c/c	3.369 kN	Fw x spacing
Shear utilisation	19.0 %	against 17.74 kN, Table C1
Planting depth for the ground resistance	835 mm in a 400 mm hole	TWf2012:01 Eq D1/D2, ground 'average' G=390 kN/m2
Deflection, EI from Table C1 (368 kNm2)	19.0 mm = h/126	delta = $w h^4 / 8EI$, cantilever post
Deflection, E mean 11,000 N/mm2	8.9 mm = h/269	same formula, I = 71.2e6 mm4

Deflection is shown both ways because the two sources give different EI. Table C1's value of 368 kNm2 gives h/126 against the h/125 marker

(19.2 mm) - inside it, but not by much. With E = 11,000 N/mm2 it is h/269. The designer must decide which applies and confirm it. A deeper

post is the only fix: 75 x 225 already has the highest EI in Table C1, and closer centres do not reduce post deflection because it depends on

the line load only.

4. CROSS-CHECK AGAINST THE VALIDATED TOOL

The same inputs were run through scripts/tw_windcheck.py, the calculator that reproduces TWf2012:01 Appendix E exactly (11 of 11

published values). Agreement is what makes the figures above usable at all.

Swind m/s mine 25.500 tool 25.500 AGREES

qp kN/m2 mine 0.468 tool 0.468 AGREES

Fw kN/m mine 1.685 tool 1.685 AGREES

Mo kNm/m mine 2.022 tool 2.022 AGREES

Mo per post kNm mine 4.043 tool 4.043 AGREES

planting m mine 0.835 tool 0.835 AGREES

post spacing limit m mine 3.113 tool 3.113 AGREES

5. THE ARRANGEMENT IT SUPPORTS

- Posts 75 x 225 C24 at 2.00 m centres, 6 in total, planted 900 mm into 400 mm diameter holes in mass concrete.

- Rails 100 x 75 C24 laid flat at 0.225, 1.200, 2.300 m centres.

- Facing 18 mm OSB/3, fixed to the outside face. Gate leaf 1.20 x 2.10 m.

- Mo per post 4.04 kNm against 6.294 kNm permissible = 64% used.

6. NOT APPLICABLE, AND WHY, SO NOBODY HAS TO ASK

- Wind IS applicable. The hoarding is external and open to the sky.
- Building Regulations do NOT govern it. They apply to the building, not the hoarding.
- Highways Act 1980 s.169 does NOT apply. The hoarding is inside the yard, not on the highway.

6. OPEN INPUTS - NOTHING IS ORDERED UNTIL THESE ARE CLOSED

1. THE SITE. vb,map comes from BS 5975-2 Figure 8 for the actual location, and the table

above shows 22 and 25 m/s, coastal and inland. Confirm the site and the designer locks

the row. TOWN/CONFIRM: the yard is between buildings, so 'town' is assumed, not proven.

2. GROUND. Classification 'average' is assumed from the tarmac surface. The geotechnical

report classifies it; if it is 'poor' the planting depth and spacing both change.

3. THE RUN LENGTH. 6.00 m front + two 1.20 m returns is scaled from the photograph.

Measure the bay and the drawings scale accordingly.

4. TARMAC REINSTATEMENT and whether the surface may be broken. Six holes, 400 x 900.

5. Whether anything SHIELDS the yard that will still be there for the whole life of the

hoarding. TWf 4.2.3(c): shielding may be demolished or removed.

A wind check is not a permit to load. Nothing is erected or loaded against this until the designer signs and the permit is issued.